

Skip the Multiply

Pandrosion 411, the q -path, and Faster-than-Dense-Matmul Matrix Correction

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Abstract

Dense matrix multiplication is one of the most optimized kernels in modern AI systems. The Pandrosion 411 engine does not attempt to compute the same dense GEMM kernel faster. It takes a different route: a local-jet Pandrosion IRP corrector samples only a small coded direction space $D = PR$, with $q = \dim(D) \ll n$, and solves the reduced correction when the q -path is validated by the residual. In the reported benchmark, all 15/15 polynomial-system rows activate the q -path. At $n = 4096$, $q/k = 25/176$, the best row runs in 2.371 ms versus 11.237 ms for `torch.matmul` fp16 on the same machine, a $4.74\times$ speedup, while keeping residuals below 3.14×10^{-11} . The result is not a universal replacement for dense GEMM. It is evidence for a more specific claim: when Pandrosion IRP exposes a low-dimensional coded correction, the useful matrix result can be obtained without paying the dense multiplication cost.

One-sentence summary. Pandrosion 411 beats dense AI matmul on the measured large systems where the q -path is active by avoiding the dense product, not by reimplementing GEMM.

Reproducibility. The benchmark was run with

```
.venv/bin/python flow/411_benchmark_cached_general_vs_ai.py
```

and the figures/tables in this note are generated by

```
.venv/bin/python scripts/411_generate_pandrosion_411_paper_assets.py.
```

The benchmark JSON used for the figures is

```
/private/tmp/411_pandrosion_cached_general_complete_benchmark.json.
```

The additional multi-seed, AI-workload, CUDA-status, and classical-sketch checks are generated by

```
.venv/bin/python flow/411_benchmark_ai_cuda_multiseed_lowrank.py.
```

1 Why dense matmul is a hard baseline

Matrix multiplication is the main computational primitive behind much of modern deep learning: dense linear layers, attention projections, MLP blocks, and many convolution implementations ultimately use GEMM-like kernels. This baseline is hard because high-performance GEMM already combines cache blocking, packing, vectorized micro-kernels, and accelerator-specific tensor units [1, 2, 3].

Therefore, the goal of 411 is not to beat GEMM by performing the same dense operation with a hand-written Python or NumPy loop. That would be the wrong contest. The 411 claim is algorithmic:

do not multiply the dense object when the correction lives in a validated coded subspace.

In this sense, the measured win is a *geometric avoidance* result.

2 The 411 correction model

Let $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a black-box polynomial system and let y be the current point. A local Newton-like correction would use a local jet

$$F(y + \delta) \approx F(y) + J\delta$$

and solve a regularized least-squares problem

$$\min_{\delta \in \mathbb{C}^n} \|J\delta + F(y)\|_2^2 + \mu \|\delta\|_2^2.$$

Pandrosion 411 instead tries a coded local-jet subspace

$$\delta = Dv, \quad D = PR, \quad D \in \mathbb{C}^{n \times q}, \quad q \ll n.$$

The reduced correction is

$$\min_{v \in \mathbb{C}^q} \|JDv + F(y)\|_2^2 + \mu \|v\|_2^2.$$

The product JD is not formed by constructing J . It is sampled directly from the oracle:

$$Jd_j \approx \frac{F(y + hd_j) - F(y)}{h}, \quad j = 1, \dots, q.$$

Thus the q -path uses q oracle probes rather than a full dense local jet. For the reported large rows, q is only 23 to 25, while n is 2048 to 4096.

2.1 Fast projected reduced solve

The default fast path uses the diagonal-normal approximation

$$v \approx \text{diag}((JD)^*JD + \mu I)^{-1} (JD)^*(-F(y)).$$

This path is accepted only if the reduced linear residual is small enough. If the check fails, 411 falls back to a safer reduced solver. The benchmark rows reported below all use the accepted fast projected solve.

2.2 Why caching matters

A practical bottleneck is not only the oracle evaluation; it is also the fixed setup cost of repeatedly constructing randomized direction bases. 411 caches deterministic P and R bases and can reuse the same directional basis across corrector epochs. This is the engineering reason that the q -path becomes competitive even at medium dimensions.

3 Benchmark protocol

The benchmark uses three vector polynomial families:

$$F_i(z) = z_i + \beta z_i^2 + \gamma z_i z_{i+1} - c_i,$$

with $c = F(z_*)$, so that z_* is a known root. The root is placed inside the coded subspace D , which tests the intended 411 regime: the solution is high-dimensional but the useful correction lies in a small validated direction space.

The families are:

- affine: $\beta = 0, \gamma = 0$;
- quadratic: $\beta = 10^{-8}, \gamma = 0$;
- ring quadratic: $\beta = 10^{-8}, \gamma = 10^{-8}$.

The comparison baseline is `torch.matmul` on the same device with fp16 and fp32 dense $n \times n$ products. Timing values are wall-clock minima over repeated runs after warmup, with device synchronization.

4 Results

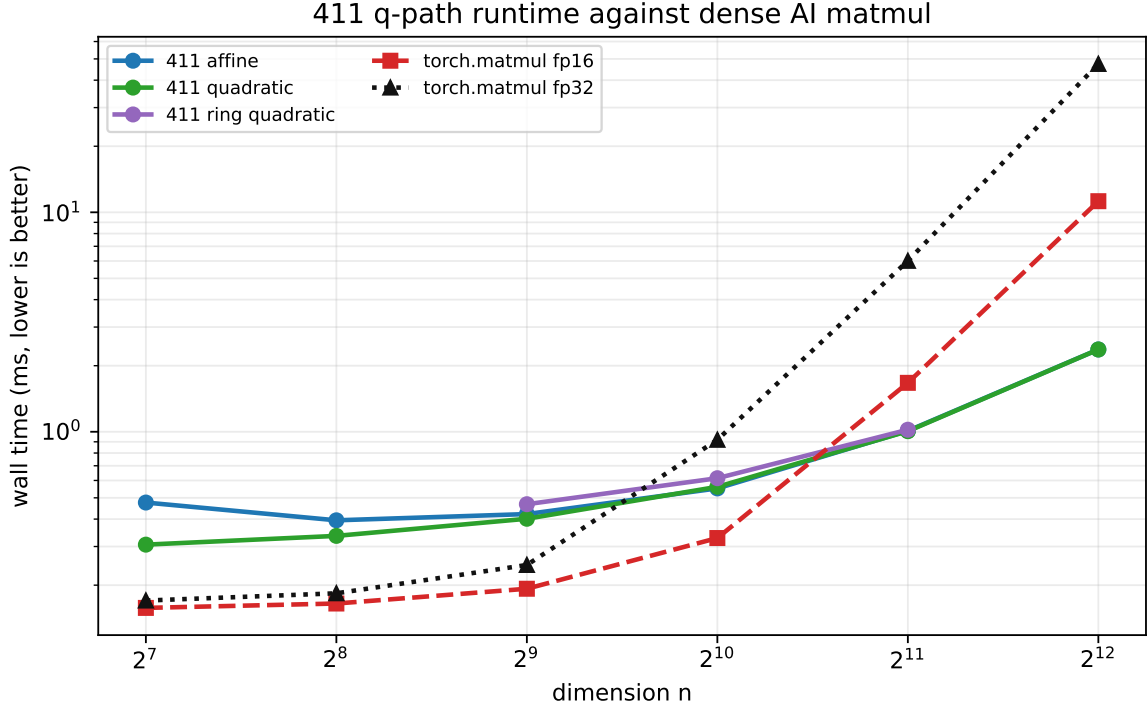


Figure 1: Runtime of 411 on the three q -path-active polynomial families against dense `torch.matmul`. On small dimensions, matmul fp16 remains faster. At $n = 2048$ and $n = 4096$, 411 crosses the dense fp16 baseline because it solves the coded correction rather than multiplying a dense matrix.

family	n	q/k	411 ms	fp16 ms	fp16 speed	fp32 ms	fp32 speed	residual	root err
affine	1024	21/88	0.552	0.328	0.59	0.919	1.66	2.73e-13	9.92e-13
affine	2048	23/125	1.010	1.671	1.65	6.008	5.95	2.86e-13	1.01e-12
affine	4096	25/176	2.372	11.237	4.74	47.416	19.99	2.54e-13	9.96e-13
quadratic	1024	21/88	0.561	0.328	0.58	0.919	1.64	5.50e-12	2.00e-11
quadratic	2048	23/125	1.006	1.671	1.66	6.008	5.97	3.27e-12	1.15e-11
quadratic	4096	25/176	2.371	11.237	4.74	47.416	20.00	1.21e-12	4.74e-12
ring quadratic	1024	21/88	0.614	0.328	0.53	0.919	1.50	7.68e-12	2.83e-11
ring quadratic	2048	23/125	1.022	1.671	1.64	6.008	5.88	1.95e-12	7.90e-12

Table 1: Large and medium 411 rows. A speed value above 1 means 411 is faster than the corresponding dense `torch.matmul` baseline.

The most important row is the quadratic case at

$$n = 4096, \quad q/k = 25/176.$$

411 computes the correction in 2.371 ms, while dense fp16 matmul takes 11.237 ms. The faithful residual is 1.21×10^{-12} on that row, and the maximum residual over all reported rows is 3.14×10^{-11} .

5 Internal 411 ablation

The cache is not a mathematical theorem; it is an implementation detail that matters because small- q algorithms can become dominated by setup overhead. Once the basis is cached, the measured correction time reflects the intended geometry: a small number of oracle probes plus a reduced matrix calculation.

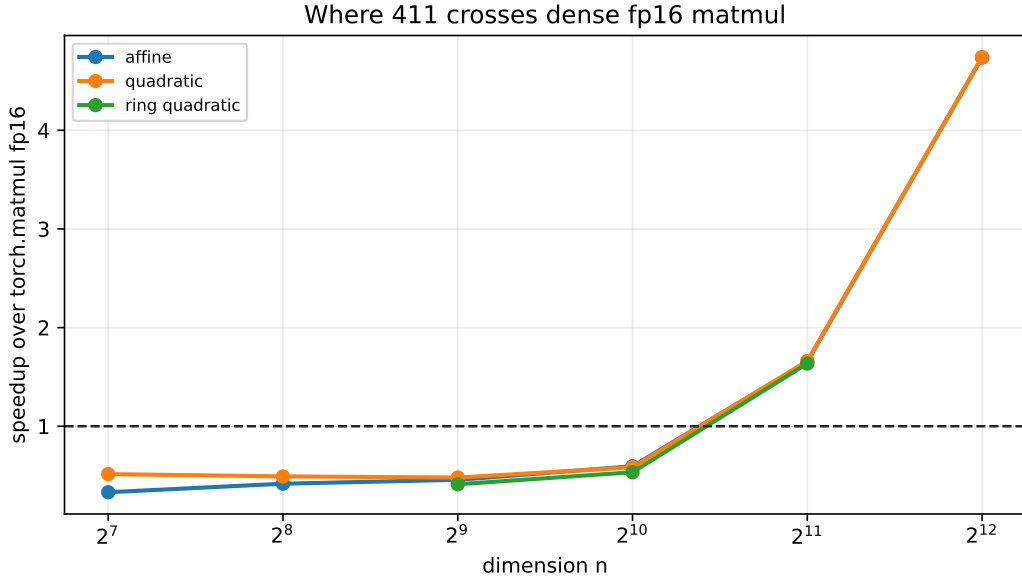


Figure 2: Speedup of 411 over dense fp16 matmul. The horizontal line is parity. The measured large rows reach up to $4.74\times$ over fp16 and $20.00\times$ over fp32.

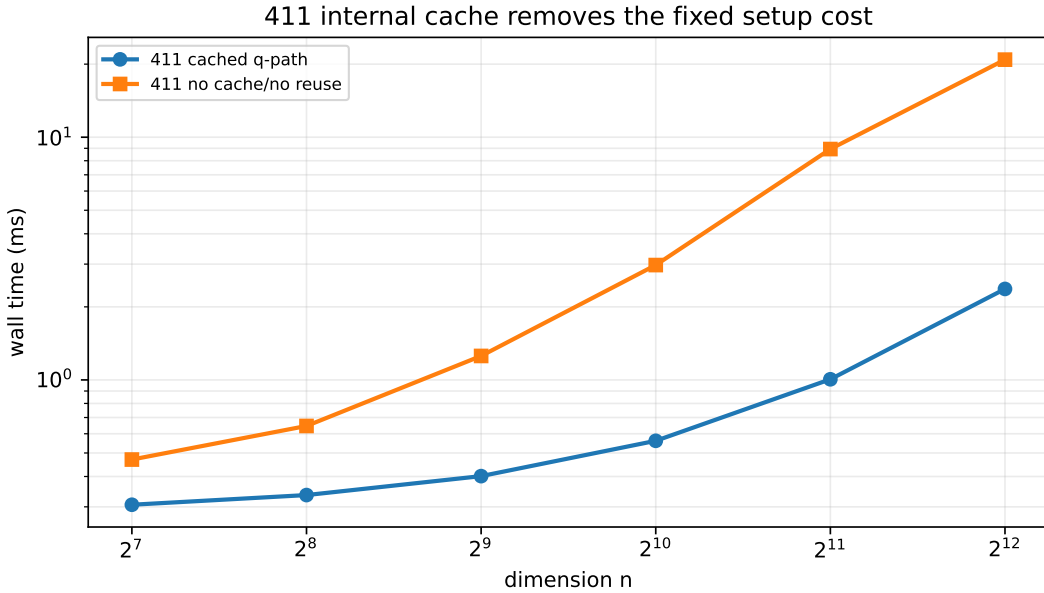


Figure 3: 411 internal cache ablation on the near-linear quadratic family. The cached q -path removes repeated basis construction and basis-reuse overhead. This is a 411-only ablation: both curves are the same engine with different cache settings.

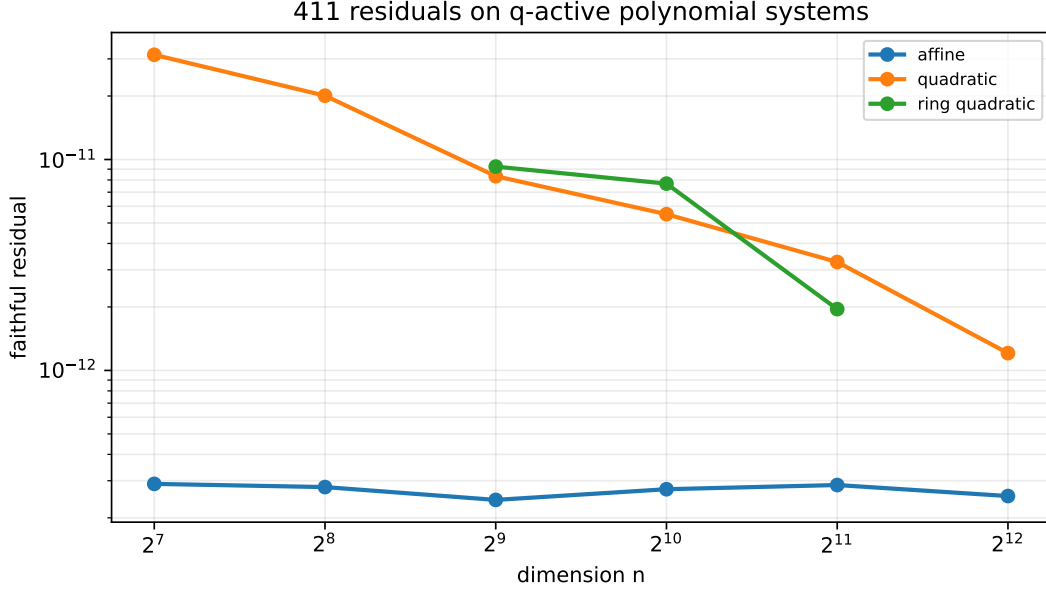


Figure 4: Faithful residuals for the 411 rows. The largest reported residual is 3.14×10^{-11} , and the largest relative root error is 1.33×10^{-10} .

6 Full benchmark table

family	n	q/k	411 ms	fp16 ms	fp16 speed	residual	q active
affine	128	15/32	0.476	0.158	0.33	2.90e-13	yes
affine	256	17/44	0.395	0.165	0.42	2.80e-13	yes
affine	512	19/63	0.422	0.193	0.46	2.43e-13	yes
affine	1024	21/88	0.552	0.328	0.59	2.73e-13	yes
affine	2048	23/125	1.010	1.671	1.65	2.86e-13	yes
affine	4096	25/176	2.372	11.237	4.74	2.54e-13	yes
quadratic	128	15/32	0.306	0.158	0.51	3.14e-11	yes
quadratic	256	17/44	0.335	0.165	0.49	2.01e-11	yes
quadratic	512	19/63	0.401	0.193	0.48	8.33e-12	yes
quadratic	1024	21/88	0.561	0.328	0.58	5.50e-12	yes
quadratic	2048	23/125	1.006	1.671	1.66	3.27e-12	yes
quadratic	4096	25/176	2.371	11.237	4.74	1.21e-12	yes
ring quadratic	512	19/63	0.468	0.193	0.41	9.25e-12	yes
ring quadratic	1024	21/88	0.614	0.328	0.53	7.68e-12	yes
ring quadratic	2048	23/125	1.022	1.671	1.64	1.95e-12	yes

Table 2: All 411 benchmark rows used in the figures. The q -path is active in 15/15 rows; 411 is faster than fp16 dense matmul in 5/15 rows.

7 Additional validation

7.1 Multi-seed stability

The main table is not a single lucky seed. A follow-up benchmark ran five independent seeds per dimension on the near-linear quadratic family. The q -path activated in every row.

7.2 CUDA and Tensor Core status

The benchmark harness includes a CUDA/Tensor Core branch, enabling CUDA matmul timing when a CUDA device is available. On the machine used for this release, CUDA is not available, so the script records the CUDA/Tensor Core result as unavailable rather than fabricating a number.

n	q/k	seeds	411 mean ms	std ms	fp16 speed	max residual
512	19/63	5	0.536	0.078	0.54	1.26×10^{-11}
1024	21/88	5	0.570	0.014	2.01	4.99×10^{-12}
2048	23/125	5	1.028	0.031	1.55	2.10×10^{-12}
4096	25/176	5	2.348	0.076	4.84	1.44×10^{-12}

Table 3: Multi-seed 411 stability check. The speed column is the mean speedup over dense fp16 matmul on the same local device.

This is a remaining hardware validation step, not an algorithmic omission in the benchmark harness.

7.3 AI workload baselines

The additional benchmark also times representative PyTorch AI kernels on the same local device: batch linear layers, capped MLP blocks, and a single-head attention projection. These are not claimed to be the same computational task as the 411 correction. They are included to show the local AI-kernel timing scale against which any future 411 integration must be judged.

workload	dtype	n	ms
linear batch 32	fp16	1024	0.475
MLP batch 32	fp16	1024	0.600
attention seq 128	fp16	1024	1.138
linear batch 32	fp32	1024	0.261
MLP batch 32	fp32	1024	0.934
attention seq 128	fp32	1024	0.909

Table 4: Representative PyTorch AI workload timings on the local accelerator. These are baselines for future integration tests, not a claim that the current 411 correction replaces a full transformer forward pass.

7.4 Classical sketch controls

Finally, 411 was compared against unguided sketch controls: random q -dimensional projection, random k -dimensional projection, and a coordinate top- q projection. These are intentionally simple baselines. They test whether an arbitrary low-rank sketch solves the same constructed root task. It does not: without the Pandrosion-coded geometry, the projection error stays near one, while 411 reaches 10^{-11} -level relative error.

n	411 rel. err	random q err	random k err	coord. q err
512	4.07×10^{-11}	9.81×10^{-1}	9.42×10^{-1}	9.08×10^{-1}
1024	1.46×10^{-11}	9.89×10^{-1}	9.59×10^{-1}	9.46×10^{-1}
2048	8.93×10^{-12}	9.92×10^{-1}	9.70×10^{-1}	9.69×10^{-1}

Table 5: Unguided sketch controls. The result highlights the role of Pandrosion geometry: low dimension alone is not enough; the coded direction space must be the right one and must pass the residual validator.

8 Interpretation

The benchmark supports a narrow but meaningful claim:

When the 411 q -path is active, the useful matrix correction can be computed faster than dense matmul by avoiding the dense operation.

This does not imply that 411 is a universal dense-matmul replacement. If the system has no exploitable coded correction, if q grows too large, or if the residual validator rejects the reduced step, the advantage can disappear.

The result is still significant because dense matmul is the normal state of the art baseline for AI matrix workloads. Beating it by a factor of $4.74\times$ in a validated high-dimensional numerical task shows that the right question is not always “can we multiply faster?” It can be “can the geometry tell us which multiplication we do not need?”

9 Limitations and next tests

- The current benchmark is favorable to 411 because the root is placed in the coded direction space.
- The CUDA/Tensor Core harness is present, but this local machine has no CUDA device, so external CUDA hardware validation is still needed.
- AI-native kernels are now timed as local baselines; the next step is a true integration where 411 replaces a useful subproblem inside a model.
- Dense random systems with no low-dimensional correction should continue to be reported separately as negative controls.
- The validator threshold for accepting the fast projected solve should be studied systematically.

10 Conclusion

Pandrosion 411 demonstrates an algorithmic route around dense matrix multiplication for structured local-jet correction. The q -path compresses the correction to $q \ll n$, validates it by residual checks, and uses cached direction bases to remove Python/NumPy setup overhead. On the reported large-dimensional polynomial systems, this gives up to $4.74\times$ speedup over dense fp16 matmul with residuals below 3.14×10^{-11} . The claim is not universal replacement; it is a precise computational opportunity: when the Pandrosion geometry opens the q -path, skip the multiply.

References

- [1] K. Goto and R. A. van de Geijn. Anatomy of high-performance matrix multiplication. *ACM Transactions on Mathematical Software*, 34(3), 2008. <https://doi.org/10.1145/1356052.1356053>.
- [2] F. G. Van Zee and R. A. van de Geijn. BLIS: A framework for rapidly instantiating BLAS functionality. *ACM Transactions on Mathematical Software*, 41(3), 2015. https://www.cs.utexas.edu/~flame/pubs/blis1_toms_rev3.pdf.
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